Name:-Siddhi Dipak Jagtap

PRN:- 123B1B147

Code:-

```
#include <iostream>
#include <stack>
#include <string>
#include <cmath>
using namespace std;

// Function to check if a character is an operand (digit)
bool isOperand(char c) {
    return (c >= '0' && c <= '9');
}

// Function to perform the operation based on the operator
int evaluate(int operand1, int operand2, char op) {
    switch(op) {
        case '+': return operand1 + operand2;
        case '-': return operand1 - operand2;
        case '*': return operand1 * operand2;
        case '/': return operand1 / operand2;
        case '^': return pow(operand1, operand2);
        default: return 0;
    }
}

// Function to evaluate the prefix expression
int evaluatePrefix(string prefix) {
    stack<int> s;
```

```
    for (int i = prefix.length() - 1; i >= 0; i--) {
        char c = prefix[i];

        // If the character is an operand, push it to the stack
        if (isOperand(c)) {
            s.push(c - '0');
        }
        // If the character is an operator, pop two operands, evaluate, and push the result
        else {
            int operand1 = s.top(); s.pop();
            int operand2 = s.top(); s.pop();
            int result = evaluate(operand1, operand2, c);
            s.push(result);
        }
    }
    return s.top();
}

int main() {
    string prefix = "-+7*45+20";
    cout << "Prefix Expression Evaluation Result: " << evaluatePrefix(prefix) << endl;

    return 0;
}
```